

Student Guide — senior / module-06- moon-phases-tithi

IKS Curriculum

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Student Workbook

Student Workbook — Moon Phases and Tithi (Senior)

This workbook is yours to write in. Use it for vocab, exercises, and reflection across all 10 days.

Vocabulary tracker

Fill in as we go:

- *candra* — _____
- *pakṣa* — _____
- *śukla pakṣa* — _____
- *kṛṣṇa pakṣa* — _____
- *tithi* — _____
- *amāvasyā* — _____
- *pūrṇimā* — _____
- *pañcāṅga* — _____

Day-by-day notes

Day 1 — The Moon and Its Shapes

What I learned:

One question I have:

Day 2 — Two Fortnights — Pakṣa

What I learned:

One question I have:

Day 3 — Fifteen Tithis per Pakṣa

What I learned:

One question I have:

Day 4 — Building a Tithi Calendar

What I learned:

One question I have:

Day 5 — Lunar Months and Festivals

What I learned:

One question I have:

Day 6 — Panchanga — Five Limbs

What I learned:

One question I have:

Day 7 — Why Eclipses Happen

What I learned:

One question I have:

Day 8 — Modern Lunar Science

What I learned:

One question I have:

Day 9 — Cross-cultural Calendars

What I learned:

One question I have:

Day 10 — Synthesis and Quiz

What I learned:

One question I have:

Final reflection

After Day 10, write a paragraph: what is the most important thing you learned in this module?

Day 1 — The Moon and Its Shapes

Module: Moon Phases and Tithi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: observe and name the four basic phases (new, first quarter, full, last quarter).

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *candra* (IAST: candra; lit. “Moon”)

Primary source (today’s reading)

From *Sūrya-siddhānta* (c. 5th–10th c. CE) — see `sources.md`.

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *10-Day Moon Observation Diary* (see `activities/`). Today’s slice: focus on the part of the activity that reinforces The Moon and Its Shapes.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 2 — Two Fortnights — Pakṣa

Module: Moon Phases and Tithi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: distinguish śukla pakṣa from kṛṣṇa pakṣa and name today’s.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See `lesson-plan.md` for any day-specific materials)

Vocabulary introduced today

- *pakṣa* (IAST: pakṣa; lit. “fortnight (half-month)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day's idea is — a number trick, a celestial pattern, a herb's identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today's idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *10-Day Moon Observation Diary* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Two Fortnights — Pakṣa.

Wrap-up (5 min)

- One-sentence exit ticket: “*What's one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 3 — Fifteen Tithis per Pakṣa

Module: Moon Phases and Tithi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: recite the 15 tithi names (pratipadā, dvitīyā, ...) and explain why a tithi isn't exactly a day.

Materials

- Whiteboard / chart paper
- Notebook, pen per student

- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *śukla pakṣa* (IAST: śukla pakṣa; lit. “waxing fortnight (new → full)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *10-Day Moon Observation Diary* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Fifteen Tithis per Pakṣa.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 4 — Building a Tithi Calendar

Module: Moon Phases and Tithi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: construct a tithi calendar for the current month using observation and computation.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *kṛṣṇa pakṣa* (IAST: kṛṣṇa pakṣa; lit. “waning fortnight (full → new)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *10-Day Moon Observation Diary* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Building a Tithi Calendar.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 5 — Lunar Months and Festivals

Module: Moon Phases and Tithi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: map the 12 lunar months (caitra, vaiśākha, ...) to major festivals.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *tithi* (IAST: tithi; lit. “lunar day (1/30 of synodic month)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *10-Day Moon Observation Diary* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Lunar Months and Festivals.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 6 — Panchanga — Five Limbs

Module: Moon Phases and Tithi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: identify the five components: tithi, vāra, nakṣatra, yoga, karaṇa.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *amāvasyā* (IAST: *amāvasyā*; lit. “new moon”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day's idea is — a number trick, a celestial pattern, a herb's identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today's idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *10-Day Moon Observation Diary* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Panchanga — Five Limbs.

Wrap-up (5 min)

- One-sentence exit ticket: “*What's one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 7 — Why Eclipses Happen

Module: Moon Phases and Tithi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: explain solar/lunar eclipses using orbital plane geometry (5° tilt of moon's orbit).

Materials

- Whiteboard / chart paper
- Notebook, pen per student

- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *pūrṇimā* (IAST: pūrṇimā; lit. “full moon”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *10-Day Moon Observation Diary* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Why Eclipses Happen.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 8 — Modern Lunar Science

Module: Moon Phases and Tithi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: phases as sun's illumination of moon; mention libration, tides.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *pañcāṅga* (IAST: pañcāṅga; lit. “five-limbed Indian calendar (tithi, vāra, nakṣatra, yoga, karaṇa)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *10-Day Moon Observation Diary* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Modern Lunar Science.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See homework.md for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 9 — Cross-cultural Calendars

Module: Moon Phases and Tithi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compare with hijri (islamic), chinese, and hebrew lunar calendars.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *candra* (IAST: candra; lit. “Moon”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.

4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *10-Day Moon Observation Diary* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Cross-cultural Calendars.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 10 — Synthesis and Quiz

Module: Moon Phases and Tithi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: final quiz and 10-day moon-diary share.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *pakṣa* (IAST: pakṣa; lit. “fortnight (half-month)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *10-Day Moon Observation Diary* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Synthesis and Quiz.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Writing Prompts

Writing Prompts — Moon Phases and Tithi (Senior)

Use these for in-class writing, homework, or assessment. Each prompt is suitable for a 300-word response.

1. In 3–4 sentences, explain Moon Phases and Tithi to a friend who hasn't taken this module.
2. Choose one Sanskrit term from this module. Define it, then describe one observation from your own life that matches.
3. Compare Moon Phases and Tithi with a similar concept from another module or from your science textbook. What's similar? Different?
4. Identify ONE claim in this module that you'd want to test. How would you test it?
5. What was the most surprising thing in this module? Why did it surprise you?
6. Pick one source cited in this module. Write 2 sentences explaining why it matters.
7. What is one common misconception about Moon Phases and Tithi? Rebut it in 2–3 sentences.
8. Imagine you are teaching this module to a younger student. What would you say first?
9. Read one primary-source passage cited in this module. Translate or paraphrase it; comment on what it adds to your understanding.
10. Identify a contemporary debate about the IKS framework presented here (e.g., is Tirthaji's mathematics genuinely Vedic? are doṣas a valid framework today?). Take a position with two pieces of evidence.

How to use

- Pick **one** prompt per day for the writing-prompts homework.
- Or assign all of them across the module, alternating between in-class and home.
- Senior students may treat any prompt as a 500-word essay topic.

Homework

Homework — Moon Phases and Tithi (Senior)

Light daily tasks. Each takes 10–15 minutes.

1. **Day 1 — The Moon and Its Shapes** · Write 2 sentences about today's concept. (~10 min)
2. **Day 2 — Two Fortnights — Pakṣa** · Write 2 sentences about today's concept. (~10 min)
3. **Day 3 — Fifteen Tithis per Pakṣa** · Write 2 sentences about today's concept. (~10 min)

4. **Day 4 — Building a Tithi Calendar** · Write 2 sentences about today's concept. (~10 min)
5. **Day 5 — Lunar Months and Festivals** · Write 2 sentences about today's concept. (~10 min)
6. **Day 6 — Panchanga — Five Limbs** · Write 2 sentences about today's concept. (~10 min)
7. **Day 7 — Why Eclipses Happen** · Write 2 sentences about today's concept. (~10 min)
8. **Day 8 — Modern Lunar Science** · Write 2 sentences about today's concept. (~10 min)
9. **Day 9 — Cross-cultural Calendars** · Write 2 sentences about today's concept. (~10 min)
10. **Day 10 — Synthesis and Quiz** · Write 2 sentences about today's concept. (~10 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.